# Project 2

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## 1 2L Neural Network

A **2-Layer Neural Network (2L)** is a basic neural network architecture consisting of an input layer and an output layer, with no hidden layers. The input layer receives the predictor variables, while the output layer generates predictions using learned weights and a bias term. Since there are no hidden layers, the model is limited to learning simple, linear relationships between the inputs and outputs.

For this project, a 2L Neural Network is implemented using **ScalaTion** and **PyTorch** to perform regression on three datasets: Airfoil Self-Noise, Auto MPG, and Concrete Compressive Strength. The goal is to evaluate and compare the predictive performance of both implementations through hyperparameter tuning and Quality of Fit (QoF) measures, including $\mathbf{R^2}$, MSE, and RMSE, for both in-sample and validation data.

### 1.1 ScalaTion

The ScalaTion implementation uses the built-in NeuralNet_2L class with an identity activation function to predict numerical values across the three datasets. Each dataset was split into 80% training and 20% validation data, with predictors standardized using training data statistics. Hyperparameter tuning was performed by testing seven learning rates and three mini-batch sizes, selecting the combination with the lowest validation MSE. The final models were trained using ScalaTion's built-in Adam optimizer for 400 epochs and evaluated using $\mathbf{R^2}$, MSE, and RMSE for both in-sample and validation data.

Table 1: ScalaTion 2L Neural Network Hyperparameter Tuning Results

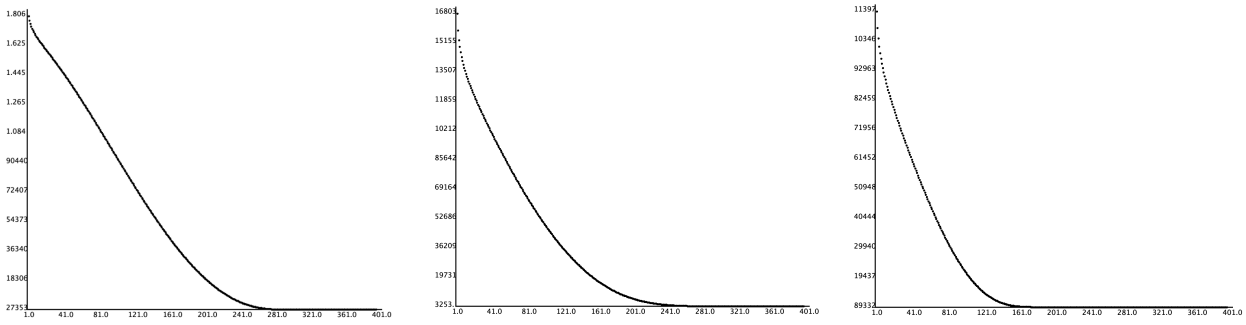
| **Dataset** | **Learning Rate** | **Batch Size** | **Validation MSE** |
|---|---|---|---|
| Airfoil | 0.30 | 16 | 21.8925 |
| Auto MPG | 0.30 | 16 | 12.6056 |
| Concrete | 0.30 | 16 | 135.7732 |

Hyperparameter tuning was performed by testing different learning rates and mini-batch sizes for each dataset. As shown in Table 1, all three datasets selected a learning rate of 0.30 and a mini-batch size of 16, as these combinations produced the lowest validation MSE among the values tested. This suggests that a higher learning rate and smaller mini-batch size were more effective for training the 2L Neural Network under the chosen settings. Although the validation MSE values differ across datasets, these values should not be directly compared because each dataset predicts a different target variable with a different numerical scale.

Table 2: ScalaTion 2L Neural Network Quality of Fit Results

| **Dataset** | **Train $R^2$** | **Validation $R^2$** | **Train MSE** | **Validation MSE** | **Train RMSE** | **Validation RMSE** |
|---|---|---|---|---|---|---|
| Airfoil | 0.5214 | 0.4880 | 22.7585 | 24.2767 | 4.7706 | 4.9271 |
| Auto MPG | 0.8248 | 0.8035 | 10.3957 | 13.0116 | 3.2242 | 3.6072 |
| Concrete | 0.6157 | 0.6076 | 108.4138 | 104.1674 | 10.4122 | 10.2062 |

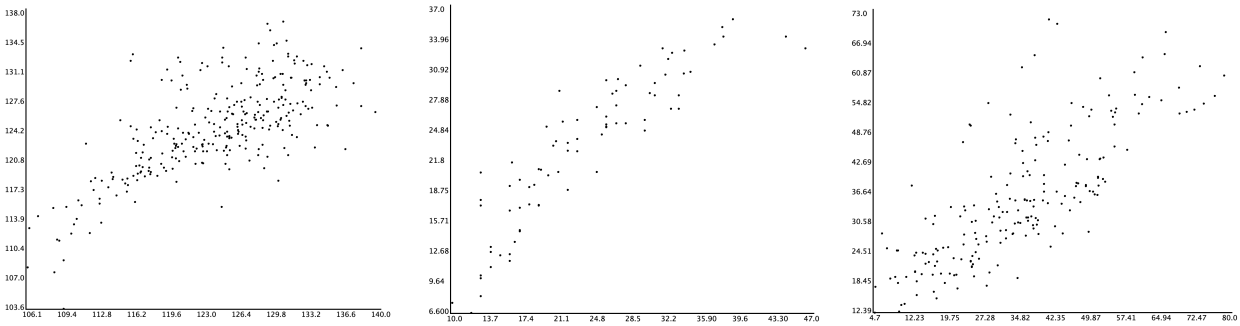
The Quality of Fit (QoF) results were evaluated using $R^2$, MSE, and RMSE for both in-sample (training) and validation data. As shown in Table 2, Auto MPG achieved the highest validation $R^2$ of 0.8035, indicating the strongest predictive performance among the three datasets based on this metric. Concrete achieved a validation $R^2$ of 0.6076, followed by Airfoil at 0.4880. The training and validation $R^2$ values were relatively close across all three datasets, suggesting no significant signs of overfitting. Overall, the results show that the 2L Neural Network performed well on Auto MPG but had more limited predictive accuracy on Airfoil and Concrete.



(a) Airfoil (b) Auto MPG (c) Concrete

Figure 1: ScalaTion 2L Neural Network Training Loss vs. Epoch for All Three Datasets

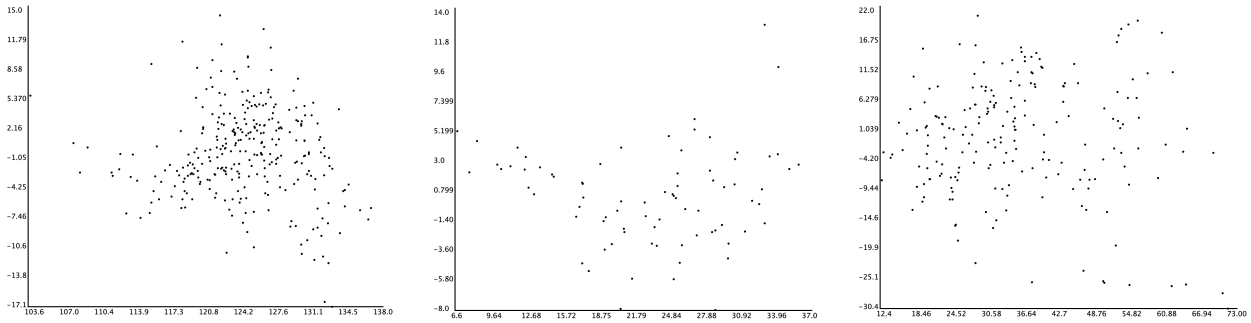
Figure 1 shows the loss vs epoch plots for the Airfoil, Auto MPG, and Concrete datasets. Across all three datasets, the training loss decreases as the number of epochs increases, indicating that the neural networks are learning and reducing prediction errors. Airfoil and Auto MPG show a gradual decrease in loss before stabilizing around 250–280 epochs, while Concrete stabilizes earlier, around 150–170 epochs. Overall, the plots suggest that all three models converged during the 400 training epochs, with minimal improvement toward the end of training.



(a) Airfoil (b) Auto MPG (c) Concrete

Figure 2: ScalaTion 2L Neural Network Validation Actual vs. Predicted Values

Figure 2 shows the validation actual vs predicted plots for the three datasets. All three plots show a positive relationship between actual and predicted values, indicating that the models capture general trends in the data. Auto MPG shows the strongest relationship, with predictions more closely following the actual values, consistent with its higher validation $R^2$ of 0.8035. Airfoil and Concrete show greater variation between actual and predicted values, suggesting lower prediction accuracy. Overall, these plots support the Quality of Fit results and demonstrate differences in model performance across the three datasets.



(a) Airfoil (b) Auto MPG (c) Concrete

Figure 3: ScalaTion 2L Neural Network Validation Residual Plots

Figure 3 shows the validation residual plots for the three datasets. The residuals are distributed above and below zero, indicating that the models both overestimate and underestimate actual values. Auto MPG shows a relatively smaller spread of residuals, consistent with its lower RMSE, while Airfoil and Concrete show greater variation and noticeable patterns in their prediction errors. Overall, the residual plots suggest that the models capture general trends but may struggle with more complex relationships in the data.

Overall, ScalaTion produced reasonable results for a 2L Neural Network across all three datasets, with Auto MPG showing the strongest predictive performance. Interestingly, Auto MPG had the fewest observations but achieved the highest validation $R^2$, while Airfoil had the most observations but the lowest validation $R^2$. This suggests that model performance depends not only on dataset size but also on the relationships between the predictor variables and the target values. Since the 2L Neural Network has no hidden layers, it may struggle to capture more complex patterns in the Airfoil and Concrete datasets. Overall, the results demonstrate that a simple neural network can perform well on certain datasets but may be limited when relationships are more complex.

## 1.2 PyTorch

The PyTorch implementation uses a 2L Neural Network with an input layer and an output layer, with no hidden layers, to predict numerical values across the three datasets. Each dataset was split into 80% training and 20% validation data, with predictors standardized using training data statistics. Hyperparameter tuning was performed by testing seven learning rates and three mini-batch sizes, selecting the combination with the lowest validation MSE. The final models were trained using the Adam optimizer for a maximum of 100 epochs, with early stopping (patience = 10) to prevent unnecessary training. Model performance was evaluated using $R^2$, MSE, and RMSE for both in-sample and validation data.

Table 3: PyTorch 2L Neural Network Hyperparameter Tuning Results

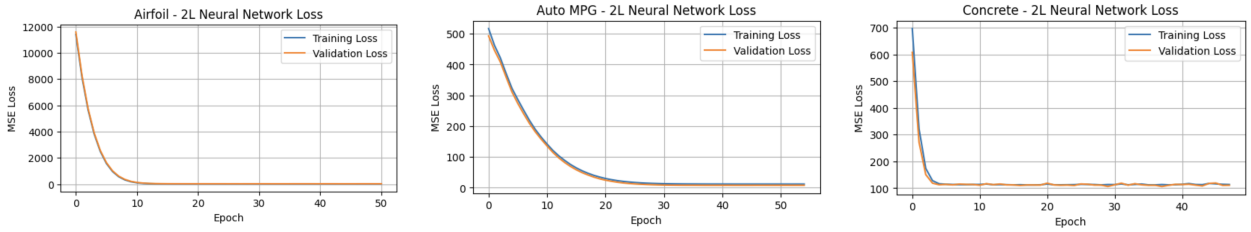
| **Dataset** | **Learning Rate** | **Batch Size** | **Validation MSE** | **Epochs** | **Best Epoch** |
|---|---|---|---|---|---|
| Airfoil | 0.30 | 16 | 31.5125 | 51 | 41 |
| Auto MPG | 0.30 | 64 | 7.8051 | 55 | 45 |
| Concrete | 0.30 | 16 | 106.1827 | 48 | 38 |

Hyperparameter tuning was performed by testing different learning rates and mini-batch sizes for each dataset, selecting the combination with the lowest validation MSE. As shown in Table 3, all three datasets selected a learning rate of 0.30, while Airfoil and Concrete selected a batch size of 16 and Auto MPG selected a batch size of 64. Early stopping was applied with a patience of 10 epochs, resulting in training ending at epochs 51, 55, and 48, respectively. The best validation performance occurred earlier, at epochs 41, 45, and 38. These results suggest that the selected hyperparameters varied slightly across datasets and that early stopping helped prevent unnecessary training once validation performance stopped improving.

Table 4: PyTorch 2L Neural Network Quality of Fit Results

| Dataset | Train $R^2$ | Validation $R^2$ | Train MSE | Validation MSE | Train RMSE | Validation RMSE |
|---|---|---|---|---|---|---|
| Airfoil | 0.4977 | 0.5528 | 23.5637 | 22.4060 | 4.8542 | 4.7335 |
| Auto MPG | 0.8255 | 0.7829 | 11.0082 | 11.0810 | 3.3179 | 3.3288 |
| Concrete | 0.6053 | 0.6236 | 112.1226 | 96.9818 | 10.5888 | 9.8479 |

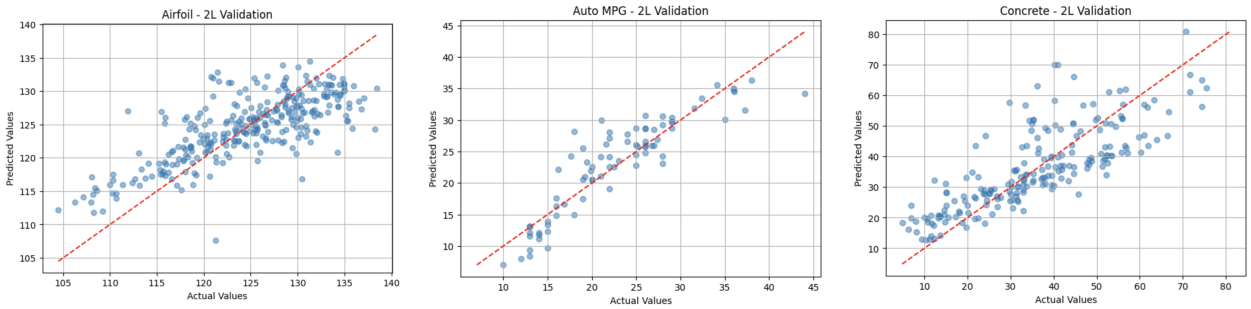
The Quality of Fit (QoF) results were evaluated using $R^2$, MSE, and RMSE for both in-sample (training) and validation data. As shown in Table 4, Auto MPG achieved the highest validation $R^2$ of 0.7829, followed by Concrete at 0.6236 and Airfoil at 0.5528. The training and validation $R^2$ values were relatively close across all three datasets, suggesting no significant signs of overfitting. Interestingly, Airfoil and Concrete achieved slightly higher validation $R^2$ values than training $R^2$ values, which can occur due to differences between the training and validation datasets. Overall, the results suggest that the 2L Neural Network performed best on Auto MPG, while Airfoil and Concrete showed more moderate predictive performance.




(a) Airfoil (b) Auto MPG (c) Concrete

Figure 4: PyTorch 2L Neural Network Training and Validation Loss vs. Epoch

Figure 4 shows the training and validation loss vs epoch plots for the Airfoil, Auto MPG, and Concrete datasets. Across all three datasets, the training and validation losses decrease as the number of epochs increases, indicating that the models are learning and reducing prediction errors. Airfoil shows a rapid decrease in loss before stabilizing around 10–15 epochs, while Auto MPG decreases more gradually and stabilizes around 25–30 epochs. Concrete shows a sharp initial decrease followed by small fluctuations in loss. Overall, the training and validation curves remain relatively close across all three datasets, suggesting that the models converged without significant signs of overfitting. Early stopping also helped prevent unnecessary training once validation performance stopped improving.

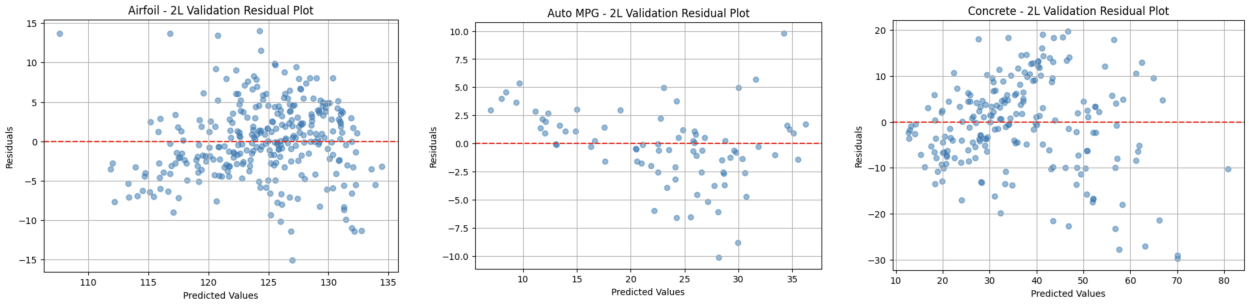



(a) Airfoil (b) Auto MPG (c) Concrete

Figure 5: PyTorch 2L Neural Network Validation Actual vs. Predicted Values

Figure 5 shows the validation actual vs predicted plots for the three datasets. The red dashed diagonal line represents perfect predictions, where predicted values equal actual values. All three datasets show a positive relationship between actual and predicted values, indicating that the models capture general trends in the data. Auto MPG shows the strongest relationship, with points generally closer to the diagonal line, consistent with its higher validation $R^2$ of 0.7829. Airfoil and Concrete show greater variation around the line,

indicating less accurate predictions. Overall, these plots support the Quality of Fit results and demonstrate that the 2L Neural Network performs differently depending on the dataset.




(a) Airfoil (b) Auto MPG (c) Concrete

Figure 6: PyTorch 2L Neural Network Validation Residual Plots

Figure 6 shows the validation residual plots for the three datasets. The red dashed horizontal line represents zero prediction error, with points above and below the line indicating underestimation and overestimation, respectively. Airfoil shows residuals scattered around zero with some noticeable variation, while Auto MPG shows a curved pattern, suggesting that the model may not fully capture certain relationships in the data. Concrete shows a wider spread of residuals, particularly at higher predicted values, indicating greater prediction uncertainty in those ranges. Overall, the plots suggest that the models capture general trends but have limitations when predicting more complex relationships.

Overall, PyTorch produced reasonable results for a 2L Neural Network across all three datasets, with Auto MPG achieving the highest validation $R^2$ of 0.7829. Interestingly, Auto MPG had the fewest observations but performed better than Airfoil and Concrete. This suggests that having more data does not always lead to better predictions, as performance also depends on the relationships between the input features and target values. Airfoil and Concrete showed moderate predictive performance, suggesting that their relationships may be more difficult for a simple neural network to capture. Since the 2L Neural Network has no hidden layers, it may struggle with more complex patterns in the data. Overall, the results show that PyTorch's 2L Neural Network can make useful predictions, but its accuracy depends on the dataset and the relationships within it.

## 1.3 Conclusion

Overall, both ScalaTion and PyTorch produced similar results when implementing the 2L Neural Network across the three datasets. Auto MPG achieved the highest validation $R^2$ in both implementations, despite having the fewest observations, suggesting that more data does not always guarantee better predictions. ScalaTion performed slightly better on Auto MPG, while PyTorch achieved higher validation $R^2$ values for Airfoil and Concrete. One important difference is that ScalaTion trained for 400 epochs, while PyTorch used a maximum of 100 epochs with early stopping (patience = 10). These different training conditions may have contributed to the differences in performance. Overall, the results show that a 2L Neural Network can make useful predictions, but its accuracy depends on the dataset and may be limited when relationships are more complex.